Cognitive-Enhancing Drugs and Their Efficacy on the Island

Stats 101b Final Project

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Note: All code is accessible at github.com/kevbaer/Stats_101b_Final_Project for following along!

1. Introduction

The professional chess governing body has a list of banned substances which includes certain “cognitive-enhancing drugs” and other stimulants, including Methamphetamine, but not caffeine. Through clinical trials, Franke et al. (2017) observed that there is an effect of cognitive-enhancing drugs on chess ability, but not on in-match performance due to issues of playing quickly enough, with caffeine showing no significant performance benefit within their study. Cappelletti et al. (2017) have noted that caffeine is known to increase perception of alertness and can induce anxiety at higher doses

In order to address these findings within a controlled environment, this experiment uses a repeated measures design to effectively evaluate whether distinct dosages of cognitive enhancers affect chess performance scores of each participant. Whether a player’s biological sex or an interaction between treatment and sex plays a role in the outcome of chess performance is what will be experimented with for this design philosophy.

2. Methods

The research was restricted to young adults aged 21-29 located on Bonne Sante Island. To maintain the data as randomly selected and unbiased as possible, households and participants were selected through a random sampling generator. Sampling through a series of cities prevents any uneven baseline populations across the Island. The final sample consisted of the first 34 eligible individuals who provided consent for the experiment. The experiment group maintained a balance across biological sex to support the integrity of the experiment.

The experiment’s design was conducted as a two-factor repeated measures design, which allowed all 34 players to go through all four treatment groups, resulting in 136 treatment blocks and 816 total chess games. The structure of the experiment is defined as follows:

1. Response Variable: Building off of this experiment, since chess performance is a key piece to analyze the effects of each treatment condition, the player’s performance in a chess game was recorded as the response variable. The score follows a standard rule: a loss yielded 0 points, a draw yielded 0.5 points, and a win yielded 1 point, establishing a maximum possible value of 6 points in the span of 6 games for each participant.Participants start off with playing chess on the white side and then transitioning to the black side after.
2. Factor 1: Consisted of four treatment groups: Control, single caffeine tablet (100 mg), double caffeine tablet (200 mg), and methamphetamine injection (10 mg).
3. Factor 2: Categorized the biological sex as Male (M) or Female (F).
4. The factor diagram for the experiment is as follows:

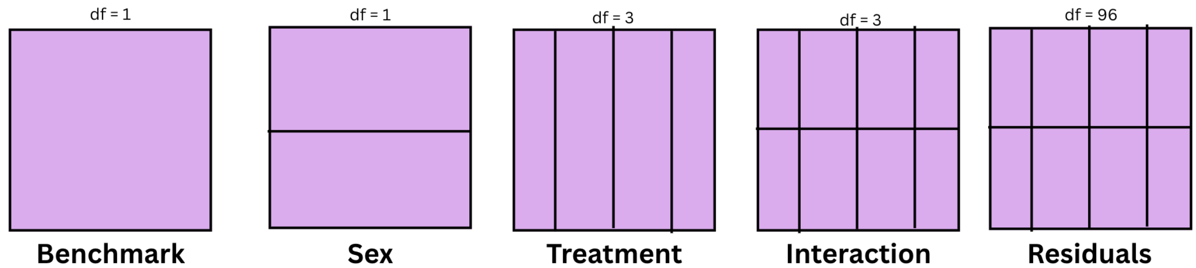


Figure 1: Factor Diagram

The experiment relied on effective administration to execute procedures and catalog player outcomes.

- Chemical treatments were provided using single caffeine tablets (100 mg), double caffeine tablets (200 mg), and methamphetamine injections (10 mg) offered by the simulation of the Island.
- The simulation provided an interactive simulation, detailing the outcome of each participant's games as well as the time.
- Sample size was determined G*Power. Inferential analysis, models of analysis for repeated measures, Tukey HSD, and residual plotting were all computed using R.

Procedures:

1. Determine the required sample size with the desired power and effect size. The goal was to have a power of 0.85, an effect size of 0.25, and a significance level of 0.05. Using G*Power and the predetermined parameters, a sample size of 24 was obtained with the trust statistical power elevated to 0.94. This minimizes the probability of committing a Type II error. However, since there were 4 treatments conducted on 6 games for 2 groups, a sample size of 34 turned out to be more effective and balanced with 6 replications per treatment condition.

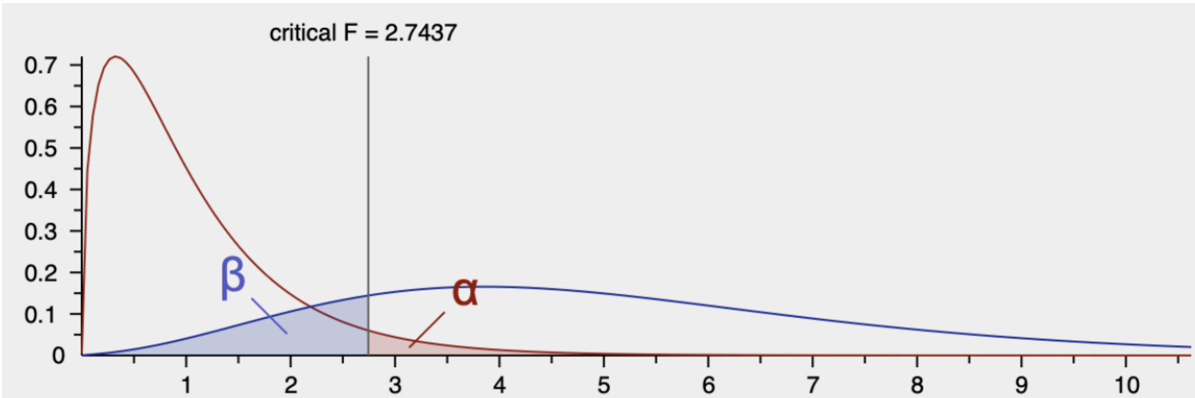


Figure 2: GPower Output

2. Design a generator that randomizes and generates a list of houses. The first 34 houses that met the age criteria (21-29), with approval of consent, were added to the study.
3. Over four consecutive days, participants were exposed to one of the four treatment levels daily to mitigate sequence and learning effects.
4. Upon receiving a stimulant or injection, participants had to wait 1 hour before proceeding with the experiment before engaging in the next chess match.
5. Start with chess on white side and record their performance based on win, loss, or draw score. Transition to the black side for the next game. This results in a total of 3 white and 3 black games back-to-back for each participant.
6. Record the results for each game and each participant.
7. Maintain the process for 4 days total. Each day should administer a different treatment group until all participants have cycled through all levels of the treatment groups.

3. Data Analysis and Results

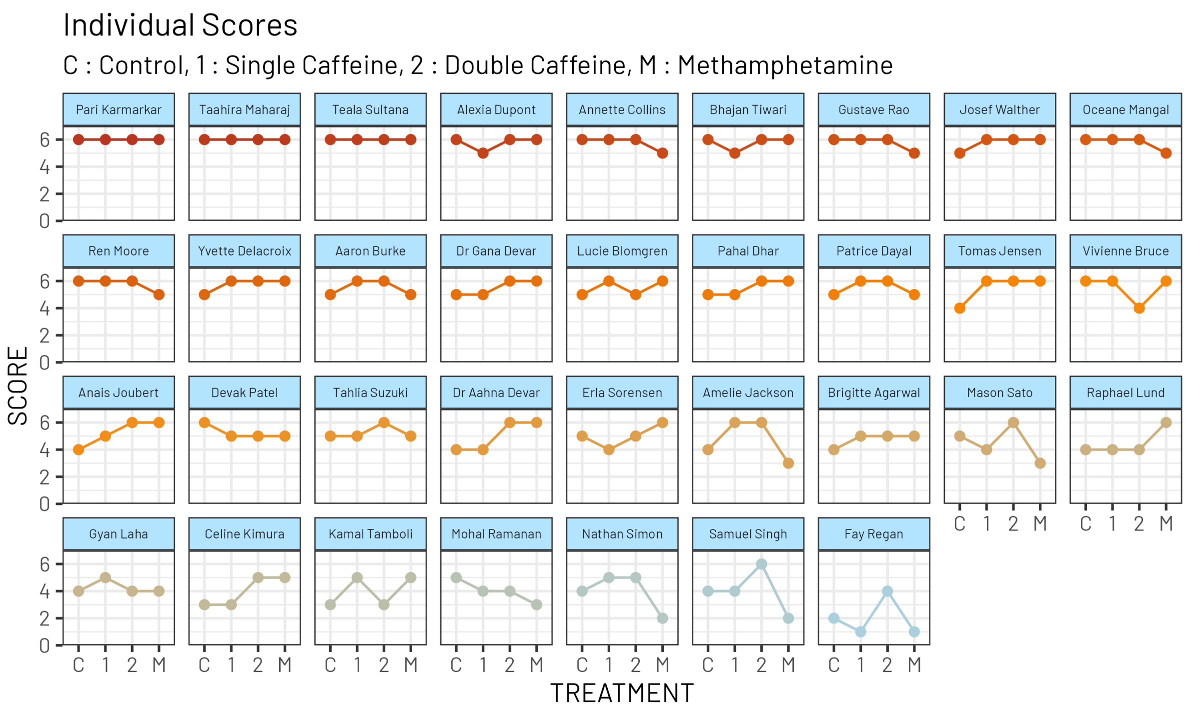
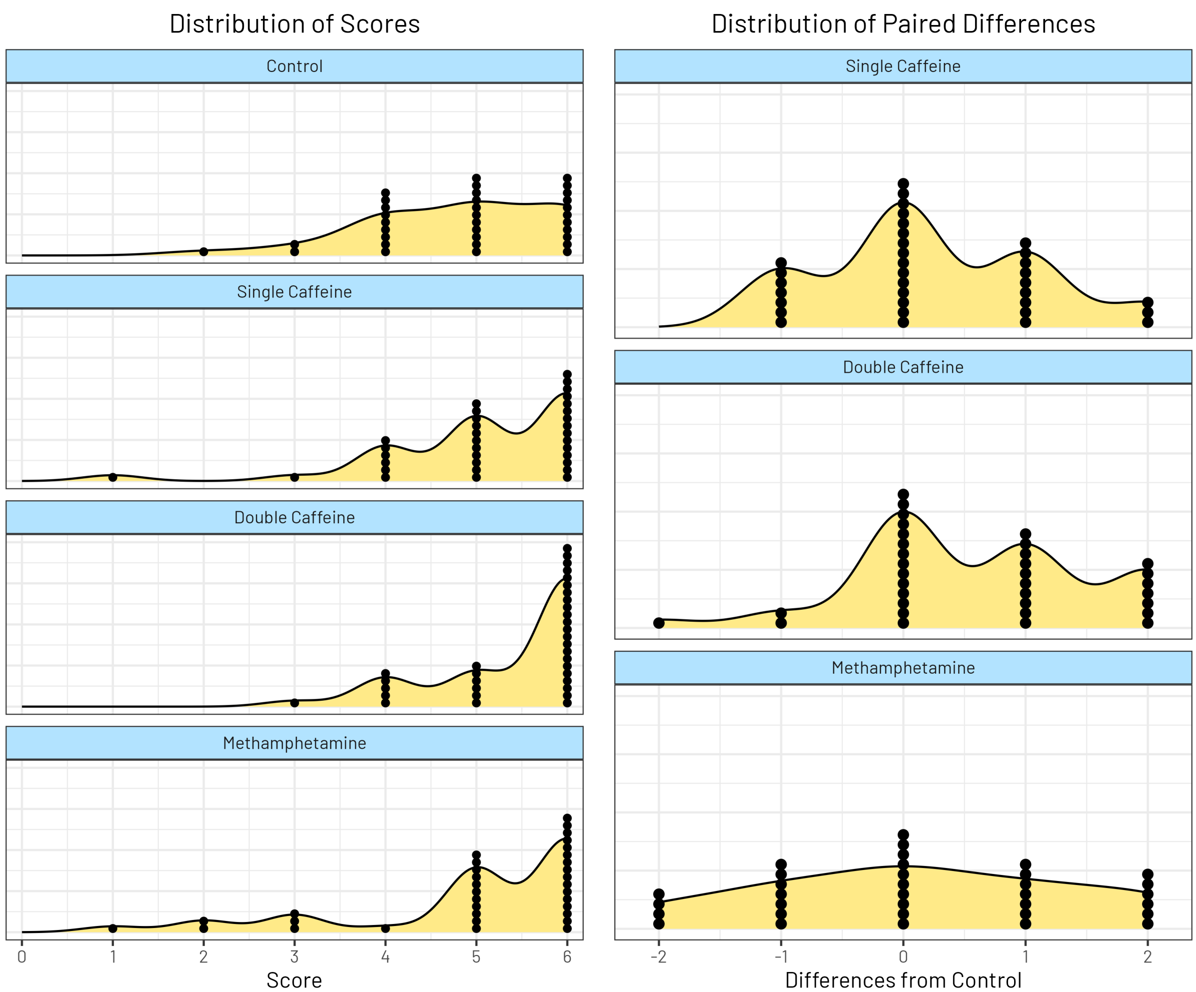


Figure 3: Individual Scores

The above figure showcases all 34 of the individual scores starting with Treatment as the x-axis and Total Scores of each treatment as the y-axis. By analyzing each participant's scores and how it trajects as they consume different treatment groups over four consecutive days, the visual interpretation of each plot reveals characteristics of the data obtained. An effective amount of participants found to have an upwards slope when transitioning to the next treatment group.

For some participants, shifting from caffeine levels to methamphetamine injection triggers an immediate downward trend.



The left plot reveals that the pattern between each treatment group hold an identical distribution. Points begin to cluster around 4 point with an increase in point density beyond that point. Control and Methamphetamine closely resemble a balanced shape as it appears that the shape begins to dip or maintain balance. Single Caffeine shows a slightly left-skewed pattern. Double Caffeine, however, displays a highly left-skewed plot, indicating an upward shift in performance.

The right figure highlights the differences from control for each stimulant tested. Single Caffeine and Methamphetamine's distribution appear tightly centered around 0, indicating that these treatments did not produce an effective change from Control for the majority of the participants. Double Caffeine, on the other hand, displays a shift towards the right, outlining that there is a positive increase when individuals transition from Control to Double Caffeine.

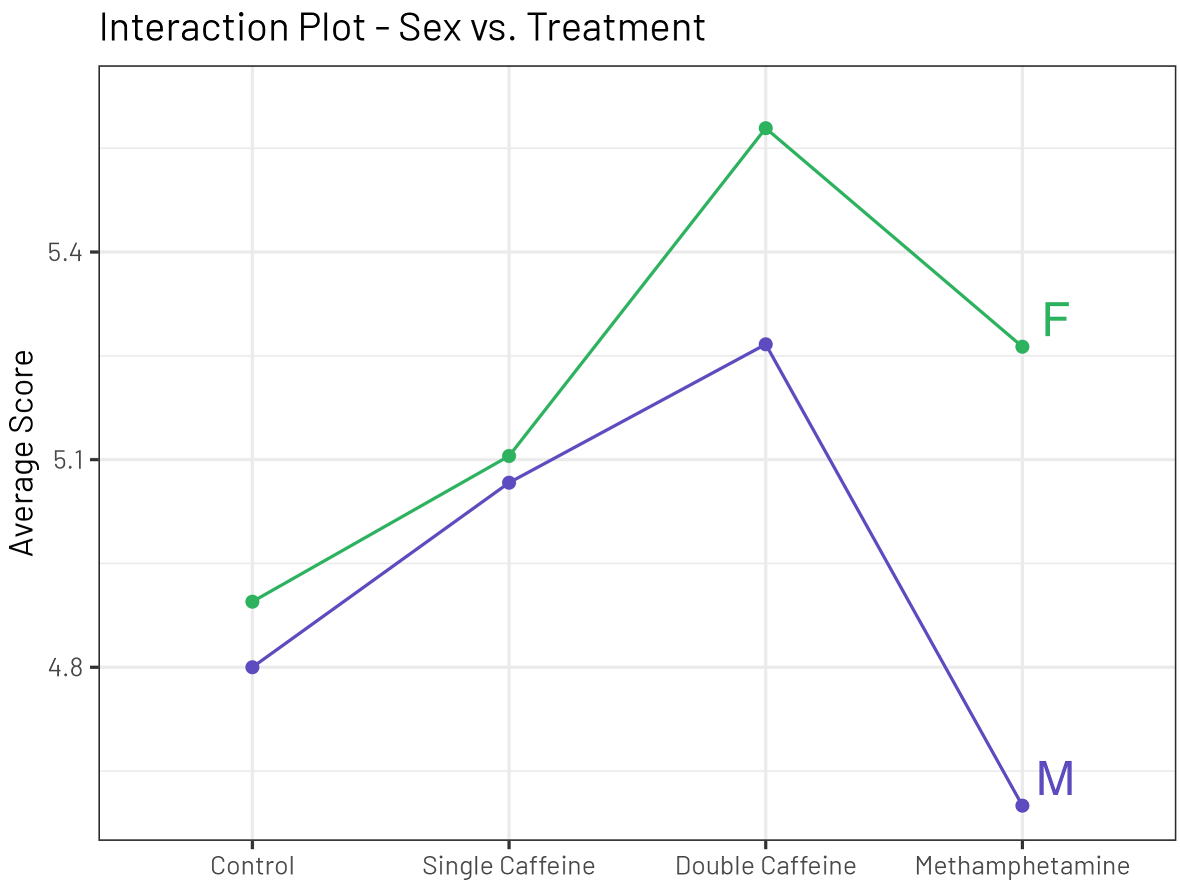


Figure 4: Interaction Plot between Treatment and Sex

The interaction plot shows that the response lines for male (M) and female (F) are somewhat parallel. However, it does not show a clear interaction between the two, indicating that there's not enough evidence that treatment differs by sex.

stratum	term	df	sumsq	meansq	statistic	p.value
NAME	SEX	1	2.58	2.576	0.869	0.3582
NAME	Residuals	32	94.86	2.965		
NAME:TREATMENT	TREATMENT	3	6.59	2.196	3.249	0.0252
NAME:TREATMENT	TREATMENT:SEX	3	2.02	0.672	0.994	0.3992
NAME:TREATMENT	Residuals	96	64.9	0.676		

Figure 5: ANOVA Table

The null hypothesis of the study states that the chess score is the same across all treatment conditions, while the alternative hypothesis states that at least one treatment differs Overall, the R^2 found was 0.62. By analyzing the ANOVA, TREATMENT is the only factor with a p-value less than alpha (0.05), therefore rejecting the null and confirming that not all treatments are equal to control.

contrast	null.value	estimate	std.error	df	statistic	adj.p.value
Single Caffeine - Control	0	0.235	0.199	99	1.18	0.6408
Double Caffeine - Control	0	0.588	0.199	99	2.95	0.0204
Methamphetamine - Control	0	0.118	0.199	99	0.59	0.9349

Figure 6: Tukey Post-Hoc Analysis

Since it was found that not all treatments are equal to control, the Post-Hoc analysis was to demonstrate that the contrast between Double Caffeine and. Control are statistically significant at p = 0.0204. With an average increase of +0.588 points over the control and a 95% confidence interval of (0.032, 1.144), Double Caffeine significantly differs from Control. Therefore, Double Caffeine improves chess performance in this population.

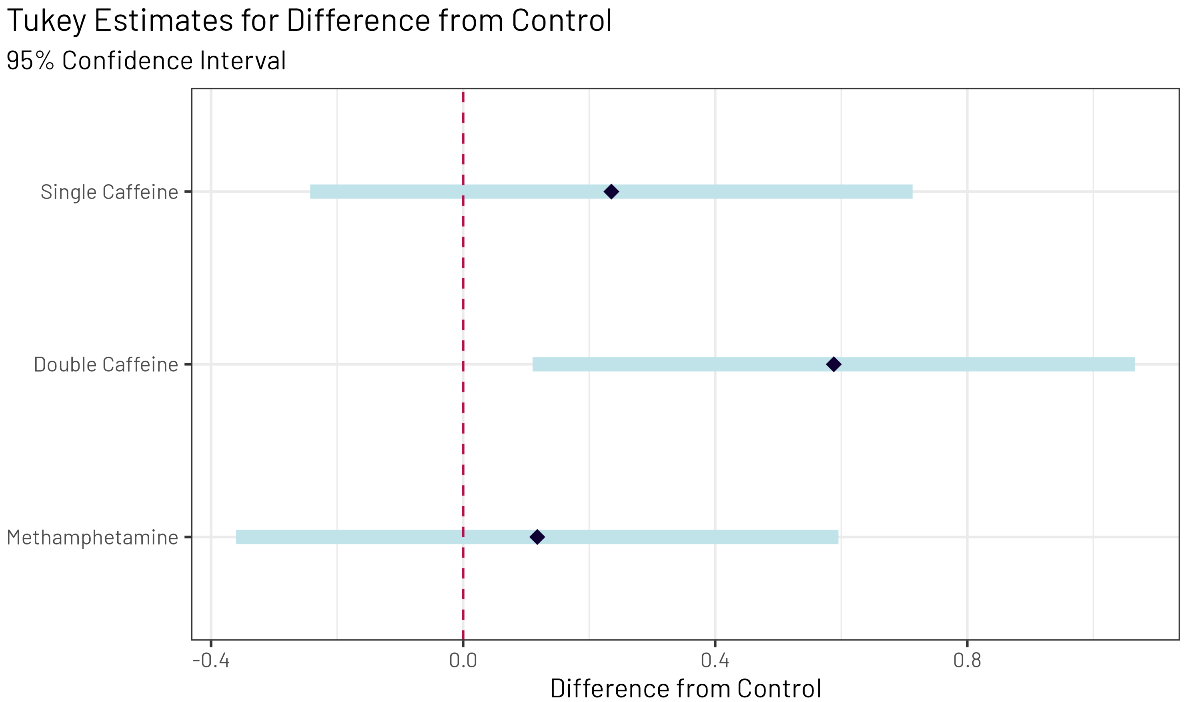


Figure 7: Tukey Post-Hoc Visualization of Differences

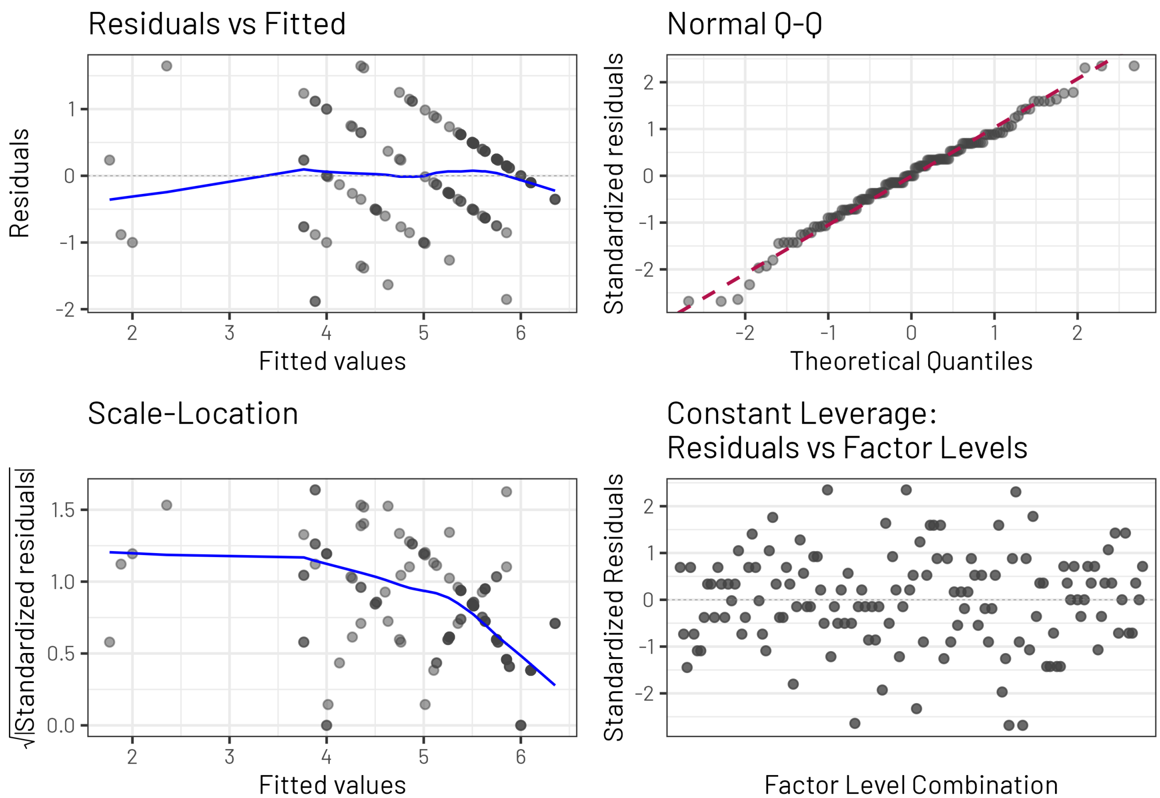


Figure 8: Residual Plot

The Residuals vs Fitted plot shows that the plots are decently spread across the plot, confirming homoscedasticity. The Q-Q Residuals plot shows that the residuals are moved tightly along a pattern, confirming that they are normally distributed. The Constant Leverage plot shows that there is a downward slope starting from left to right, suggesting that residual variance decreases as fitted values increase. The Constant Leverage plot shows that the residual spread looks reasonably consistent across all groups.

4. Conclusion

The primary goal of this study was to evaluate and determine if and how cognitive performance can be impacted by stimulants. By experimenting with different doses of caffeine tablets and the injection of methamphetamine, there is a finding in how it can be analyzed through the weight of several models of ANOVA fittings. By implementing a two-way repeated measures ANOVA design, the difference between the control group of the experiment and the treatment group allowed for an effective evaluation of which treatment group, specifically, can enhance a participant's ability to perform better in chess. Although sex did not provide an effect on treatment difference, the results provide evidence that the cognitive performances are highly dose-dependent.

When analyzing the interaction plots of each participant, majority showed an increase in chess performance when shifting to a higher dose of caffeine tablets, while some showed a slight decrease when shifting to methametamine.

Using Tukey HSD to provide a Post-Hoc Comparison between control and treatment, Double Caffeine (200 mg) was found to be the only treatment level to be statistically significant, in terms of increase from the control group. By achieving a mean score of +0.588 and a p-value of 0.0204, the table outlines that Double Caffeine improves chess performance significantly within the population measured. In addition, by analyzing the table, it also shows that methamphetamine is the least effective treatment with a p-value of 0.9349 and only a mean score of +0.118. Even in the Distribution of Paired Differences plot, methamphetamine performance appears close to uniformly shaped.

By looking at the interaction plot, the interaction between males (M) and females (F) does not appear to have a perfect parallel shape, indicating that there's not enough evidence to conclude that treatment differs by sex. In Table 1, sex only appears to have a p-value of 0.3582, which proves that sex wouldn't make a differece to the treatment outcomes of each participant.

To conclude, this study demonstrates that cognitive enhancement during these cognitive performances does enhance the effectiveness of its outcome. Specifically, double caffeine (200 mg) shows a sustainable evidence that it is the most significant in terms of improving chess performance. Although this study benefits from a balanced repeated measures design, further studies can enhance on how cognitive-enhancing drugs or other stimulants can enhance other cognitive performances to further improve on this study. Future research could further experiment whether drugs can enhance other cognitive performances such as SAT or Sudoku or if caffeine can affect how long it takes to complete a chess game. Age groups can be better reiterated through experimenting with different age groups in the future.

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